

Towards Systematic Evaluation of de Sitter Correlators via Generalized Integration-By-Parts Relations

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ABSTRACT: We generalize Integration-By-Parts (IBP) and differential equations methods to de Sitter correlators related to inflation. While massive correlators in de Sitter spacetime are usually regarded as highly intricate, we find they have remarkably hidden concise structures from the perspective of IBP. We find the factorization of the IBP relations of each vertex integral family corresponding to $d\tau_i$ integration. Furthermore, with a smart construction of master integrals, the universal formulas for iterative reduction and $d \log$ -form differential equations of arbitrary vertex integral family are presented and proved. These formulas dominate all tree-level de Sitter correlators and play a kernel role at the loop-level as well.

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1 Introduction

The quantum field theory in curved spacetime has a rich history and holds paramount significance in various physical applications. As the maximally symmetric spacetime, de Sitter (dS) spacetime as well as anti-de Sitter (AdS) spacetime should be the simplest curved spacetime to study. The quantum field theory in dS and AdS has gained large interest among researchers. In the phenomenology of particle physics and cosmology, correlators in dS spacetime play a crucial role in the study of inflationary cosmologies and possible signals of new physics, known as the cosmological collider program [1–3]. These correlators are then related to experimental observations [4] such as Cosmic Microwave Background (CMB) radiation and Large-Scale Structure (LSS). Utilizing the in-in formalism [5] in cosmology [6, 7] and its diagrammatic approach, the Schwinger-Keldysh (SK) path integral [5, 8, 9], one can derive Feynman-like diagrams and rules for n -point correlators [10]. The computation and analysis of these inflation correlators is an ongoing research topic. People have considered this problem from many aspects such as cosmological bootstrap [11–21], cutting rule [22–28], Mellin-Barnes (MB) space [29–34], partial MB transformation [35–38], integrand issues [39–45]. Nevertheless, this topic, to the best of our current knowledge, has not been systematically solved and comprehended.

On the other side, the methods developed for computing amplitudes in flat spacetime are well-established in last two decades. Similar methods have recently been applied for dS correlators. For example, MB transformation has been applied in both flat [46, 47]

and dS [29–38]. In dS case, people find the (high-order) differential equation of correlator and solve it in terms of hypergeometric functions [37]. Meanwhile, in flat space, similar method also has been considered, using the more systematic mathematical tool GKZ [48–52]. Thus, examining the distinct features of amplitudes in flat spacetime and correlators in dS spacetime and drawing insights from the computation methods for flat spacetime could provide guidance for developing computation methods of dS correlators.

In the beginning, let’s highlight the main difference between these two cases. The integrals appearing in flat amplitude take the form

$$\int_{\mathcal{C}} \prod_i P_i(\mathbf{z})^{\alpha_i} d\mathbf{z}, \quad (1.1)$$

where P_i is polynomial and the integration region $\mathcal{C}$ is properly chosen. For example, the conventional form of Feynman integral take this form, with $P_i = (l_i + q_i)^2 - m_i^2$, as well as their corresponding Feynman parametrization or Baikov representation [53]. On the contrary, the integrals appearing in dS correlator takes the form

$$\int_{\mathcal{C}} \prod_i P_i(\mathbf{z})^{\alpha_i} \prod_j F_j(\mathbf{z})^{\beta_j} d\mathbf{z}, \quad (1.2)$$

where F_i are functions related to Hankel functions and step functions [10]. This poses substantial challenges in the computation of such dS correlators. Successful experience for higher precision computations of amplitudes in flat space in the past decade leads us to the Integration-By-Parts (IBP) reduction [54] and (first-order) Differential Equations (DE) [55–58] obtained by IBP. These are what we want to introduce to dS correlators. The IBP method establishes linear relations among different integrals within a integral family by integrating total differential with zero boundary. For example, for the integral family

$$G(a) \equiv \int d^d l \frac{1}{(l^2 - m^2)^a} \quad (1.3)$$

the IBP relation gives

$$0 = \int d^d l \frac{\partial}{\partial l} \cdot l \frac{1}{(l^2 - m^2)^a} \rightarrow (d - 2a)G(a) - 2am^2 G(a + 1) = 0. \quad (1.4)$$

With such relations, one can reduce any integral in the integral family into a linear combination of a selected finite set of integrals (called Master Integrals (MI)), which are the only part need to be really calculated. If the partial derivatives of the master integrals still belong to the same integral family, reducing them back to the MI yields a system of first-order differential equations (DE) of the MI. With the boundary condition, one could solve them analytically or numerically. Moreover, when people encounter the same integral family in another scattering process, redundant computations could be avoid. The introduction of the canonical differential equation (CDE) method in [59] has significantly facilitated the advancement of high-loop analytical calculations, whose DE are d log-form. Then, numerical DE methods gained attention and generalized power series expansions

[60–62] and Auxiliary Mass Flow (AMFlow) were developed [63–66]. Meanwhile, automation packages for IBP [67–72] and numerical DE [73–75] have been continuously refined and widely applied in recent years and provided immense convenience. Due to all these developments, IBP and DE have become prioritized tools especially for computing the most complicated amplitudes. For instance, the recent result of a two-loop five-point process [76] is obtained using IBP, numerical DE and the related packages. IBP and DE are used to be applied on the integrals take the expression (1.1). This also implies that if same forms of integrands are identified in other physical scenarios, the computational techniques developed for flat spacetime Feynman integrals may be directly applied to those situations. For instance, in dS correlator, calculations are comparatively more tractable in the case of conformally coupled scalar due to their integrals also take the form (1.1), and IBP and DE have been studied [77, 78] in this case recently.

In this paper, the power and significant success of IBP and DE in flat spacetime have motivated us to promote their application to dS integrals with the form (1.2). This extension aims to develop a systematic, efficient, and automatic technique for computing general (not only massless or conformally coupled) dS correlators. It provides also a new perspective and tool to analyze the structure and properties of dS correlator.

The structure of the paper is following. In section 2, we review the Feynman rule of the dS correlator, then construct IBP of this case. In section 3.1, we indicate that each $d\tau_i$'s IBP are factorized, which leads to great simplification of the computation. For the so-called vertex integral family (with respect to integration of each τ_i), we have constructed a smart set of bases, which will lead to further significant simplifications in the iterative reduction and DE of this integral family. In section 3.2, we give a pedagogical example of iteratively reducing 1-fold Hankel function vertex integral family, to sketch the features of general cases. In section 3.3, we find it coincidentally that the DE of the MI we construct automatically are d log-form. Subsequently, we systematically present and prove the universal formula of iterative reduction for arbitrary n-fold vertex integral families in section 3.4 and its d log-form DE in section 3.5. In section 3.6, we quickly cover the simplest massless case. In section 3.7, we give discussion of IBP and DE when they involving remaining terms come from two types of propagator $G_{\pm\pm}$ and belong to the so-called sub-sector. Finally, a brief summary and discussions are presented in the section 4.

2 Generalized IBP relations of dS correlators

In this section, we review the Feynman rules of in-in formalism dS correlators in cosmology and inflation. We will find dS correlators take the form of (1.2) and determine what terms and functions could appear as P_i and F_j . Then we will generalize the IBP method to this case. For simplifying the discussion without losing key features, we focus on scalar field theory.

Following the standard notation in the inflation physics, we choose the metric with the so called conformal time τ

$$ds^2 = a^2(\tau)(-d\tau^2 + d\mathbf{x}^2), \quad (2.1)$$

where the scale factor is chosen to be $a(\tau) = 1/(-H\tau)$ with H the Hubble parameter. Feynman rules of dS correlators (see [10] for review) are similar to the case of flat space, i.e., with three basic building blocks: external wave functions, propagators and vertexes. For simplicity, let us take scalar field as an example. The mode expansion of field operator is

$$\varphi_a(\tau, \mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left[u_a(\tau, \mathbf{k}) b_a(\mathbf{k}) + u_a^*(\tau, -\mathbf{k}) b_a^\dagger(-\mathbf{k}) \right] e^{i\mathbf{k}\cdot\mathbf{x}}, \quad (2.2)$$

where the wave function $u_a(\tau, \mathbf{k})$ satisfies the equation of motion

$$u_a''(\tau, \mathbf{k}) - \frac{2}{\tau} u_a'(\tau, \mathbf{k}) + \left(\mathbf{k}^2 + \frac{m_a^2}{H^2\tau^2} \right) u_a(\tau, \mathbf{k}) = 0 \quad (2.3)$$

with m_a the mass of the field. The solution is given by

$$u(\tau; k) = -i \frac{\sqrt{\pi}}{2} e^{i\pi(\nu/2+1/4)} H^{(d-1)/2}(-\tau)^{d/2} H_\nu^{(1)}(-k\tau). \quad (2.4)$$

where $H_\nu^{(1)}(-k\tau)$ is the Hankel function and other parameters are $k = |\mathbf{k}|$, $v = \sqrt{\frac{d^2}{4} - \frac{m^2}{H^2}}$ and $d = 3$. The Hankel functions satisfy¹

$$\begin{aligned} \partial_\tau^2 H_\nu^{(1,2)}(-k\tau) + \frac{1}{\tau} \partial_\tau H_\nu^{(1,2)}(-k\tau) + \left(k^2 - \frac{\nu^2}{H^2\tau^2} \right) H_\nu^{(1,2)}(-k\tau) &= 0, \\ H_\nu^{(1)}(-k\tau) &= \left(H_{\nu^*}^{(2)}(-k^*\tau^*) \right)^*, \\ \tau \rightarrow 0_+ : H_\nu^{(1)}(-k\tau) &\sim -\frac{i2^\nu \Gamma[\nu]}{\pi} (-k\tau)^{-\nu}, \text{ for } \text{Re}[\nu] > 0 \\ \tau \rightarrow +\infty : H_\nu^{(1)}(-k\tau) &\sim \sqrt{\frac{2}{\pi}} (-k\tau)^{-\frac{1}{2}} e^{-ik\tau - i\pi(\nu/2+1/4)} \end{aligned} \quad (2.5)$$

Using (2.2) we can get various propagators. The bulk-to-bulk propagators are given by

$$\begin{aligned} G_>(k; \tau_1, \tau_2) &\equiv u(\tau_1, k) u^*(\tau_2, k), \\ G_<(k; \tau_1, \tau_2) &\equiv u^*(\tau_1, k) u(\tau_2, k), \\ G_{++}(k; \tau_1, \tau_2) &= G_>(k; \tau_1, \tau_2) \theta(\tau_1 - \tau_2) + G_<(k; \tau_1, \tau_2) \theta(\tau_2 - \tau_1), \\ G_{+-}(k; \tau_1, \tau_2) &= G_<(k; \tau_1, \tau_2), \\ G_{-+}(k; \tau_1, \tau_2) &= G_>(k; \tau_1, \tau_2), \\ G_{--}(k; \tau_1, \tau_2) &= G_<(k; \tau_1, \tau_2) \theta(\tau_1 - \tau_2) + G_>(k; \tau_1, \tau_2) \theta(\tau_2 - \tau_1), \end{aligned} \quad (2.6)$$

¹The definition of $H_\nu^{(1,2)}$ in textbook is $H_\nu^{(1)}(z) = J_\nu(z) + iY_\nu(z)$ and $H_\nu^{(2)}(z) = J_\nu(z) - iY_\nu(z)$. When z, ν are real number, it is obviously that $H_\nu^{(2)}(z) = (H_\nu^{(1)}(z))^*$. The second line of (2.5) is a generalization of above result when z, ν are complex number.

The bulk-to-boundary propagators are non-vanish only when the field is massless

$$\begin{aligned} G_+(k; \tau) &\equiv G_{+\pm}(k; \tau_1, 0) = \frac{H^2}{2k^3}(1 - ik\tau)e^{+ik\tau}, \\ G_-(k; \tau) &\equiv G_{-\pm}(k; \tau_1, 0) = \frac{H^2}{2k^3}(1 + ik\tau)e^{-ik\tau}. \end{aligned} \quad (2.8)$$

It is necessary to notice that when people consider the asymptotic past $\tau \rightarrow -\infty$, $i\epsilon$ -prescription [10] should be taken into consideration for good behavior. Practically, it is equivalent to attach an external factor $e^{\epsilon k\tau}$ to $G_>$ and $G_<$ making propagators to be exponentially suppressed to zero when $\tau \rightarrow -\infty$. For Feynman rules of vertices, some examples are listed as follows:

$$\begin{aligned} -\frac{\lambda}{24}a^4(\tau)\varphi^4 &\rightarrow \mp i\lambda \int_{-\infty}^0 d\tau a^4(\tau) \cdots \\ -\frac{\lambda}{6}a^2(\tau)\varphi(\partial_i\varphi)(\partial_i\varphi) &\rightarrow \pm \frac{i\lambda}{3}(k_{12} + k_{23} + k_{13}) \int_{-\infty}^0 d\tau a^2(\tau) \cdots, \quad k_{ij} \equiv \mathbf{k}_i \cdot \mathbf{k}_j \\ -\frac{\lambda}{6}a^2(\tau)\varphi\varphi'^2 &\rightarrow \mp \frac{i\lambda}{3} \int_{-\infty}^0 d\tau a^2(\tau) G_{+a_1}(k_1; \tau, \tau_1) \prod_{i=2,3} [\partial_\tau G_{+a_i}(k_i; \tau, \tau_i)] \\ &\quad + 2 \text{ permutations}, \end{aligned} \quad (2.9)$$

From above discussions, one can see that dS correlators are indeed the form of (1.2) with following correspondence:

$$\begin{aligned} \int_{\mathcal{C}} dz &= \int_{-\infty}^0 d\tau_i \int_{-\infty}^{\infty} d\mathbf{l}_i, \quad \mathbf{l}_i \text{ for loop momentum,} \\ P_i &= \tau_j, \quad \text{polynomial of loop momentum,} \\ F_i &= e^{ik\tau}, \quad H_\nu^{(1,2)}(-k\tau), \quad \partial_\tau H_\nu^{(1,2)}(-k\tau), \quad \theta(\tau_j - \tau_k). \end{aligned} \quad (2.10)$$

We call a integral family defined by (2.10) a **dS integral family**.

To apply IBP method, we need to check the dS integral family satisfying the proper boundary conditions. For the loop integration, just like the flat space, we will use the dimension regularization, thus the IBP relations involving $\mathbf{l}_i$ satisfy automatically. The new feature is the conformal time integration. First with the $i\epsilon$ -prescription, the integrands equal zero at the boundary $\tau_i = -\infty$. For the boundary $\tau_i = 0$, one could add a regulator $\tau_i^{\delta_i}$ if it is divergence, then it is again zero. The regulator for the divergent case is necessary for the IBP method to apply, since only with it the integration is well defined. Physical information is coded at the series expansion of δ (just like dimension regulator ϵ for flat space), for example, $1/\delta$ term in the expansion. In IBP-based differential equations method for flat spacetime, obtaining result in the expansion of regulator is also a usual practice. Since $a(\tau) \sim \frac{1}{\tau}$, power function of τ automatically appears in the integrand. With these considerations, we conclude that the IBP method can be applied to dS integral family. Let

us do one simple example

$$0 = \int_{\mathcal{C}} d \left(\tau^{\nu_0} H_{\nu}^{(1)}(-k\tau) \right), \quad (2.11)$$

gives

$$0 = \nu_0 \int_{\mathcal{C}} \tau^{\nu_0-1} H_{\nu}^{(1)}(-k\tau) d\tau + \int_{\mathcal{C}} \tau^{\nu_0} \partial_{\tau} H_{\nu}^{(1)}(-k\tau) d\tau, \quad (2.12)$$

One may notice that applying the derivative operator to $\partial_{\tau} H_{\nu}^{(1,2)}(-k\tau)$ will give $\partial_{\tau}^2 H_{\nu}^{(1,2)}(-k\tau)$, which is not in the dS integral family (2.10). However, using the relation (2.5) we can express $\partial_{\tau}^2 H_{\nu}^{(1,2)}(-k\tau)$ as the linear combination of $H_{\nu}^{(1,2)}(-k\tau)$ and $\partial_{\tau} H_{\nu}^{(1,2)}(-k\tau)$. For ∂_{k_i} , similar things also happen. For θ -function, the derivative leads to δ -function which can be easily integrated out. Overall, IBP relations obtained from a defined dS integrals family will be a closed set of equations, which can be solved systematically.

3 IBP of $d\tau_i$: vertex integral family

The integral family of $d\tau_i$ plays a key role in IBP of dS integrals and we will focus on its special properties in this section.

3.1 vertex integral family and factorization of $d\tau_i$ IBP

Before going to details, let us notice that from (2.10) for most terms, (at most times) different τ_i 's are factorized. There is only one exception, i.e., the $\theta(\tau_i - \tau_j)$, which comes from G_{++} and G_{--} . Thus we define a **vertex integral family** corresponding to these good integrands without θ -function:

$$\begin{aligned} V(\nu_0, a_1, a_2, \dots, a_n) &= V(\nu_0, \mathbf{a}) = \int \hat{V}(\nu_0, \mathbf{a}; \tau) d\tau \\ &\equiv \int_{-\infty}^0 \tau^{\nu_0} e^{ik_0\tau} \prod_i h(\nu_i, a_i; -k_i\tau) d\tau, \quad h \equiv h^{(1,2)}, \quad a_i = 0, 1, \\ h^{(1 \text{ or } 2)}(\nu, 0; -k\tau) &\equiv (-k\tau)^{-\nu} H_{\nu}^{(1)}(-k\tau) \text{ (or } H_{\nu}^{(2)}) \propto \tau^{-\frac{3}{2}-\nu} u \text{ (or } u^*), \\ h^{(1,2)}(\nu, 1; -k\tau) &\equiv -\frac{1}{k} \partial_{\tau} h^{(1,2)}(\nu, 0; -k\tau). \end{aligned} \quad (3.1)$$

In the definition we have omitted the dependence on k_0, k_i, ν_i for simplicity. We introduce also a factors $(-k\tau)^{-\nu}$ in front of Hankel function to the definition of h . This factor is one of the most important construction in this paper. One can check that the differential equation of h

$$h''(\nu, 0; -k\tau) + \frac{1}{\tau}(2\nu + 1)h'(\nu, 0; -k\tau) + k^2 h(\nu, 0; -k\tau) = 0, \quad (3.2)$$

can be rewritten as

$$\partial_{\tilde{\tau}}^2 h(\nu, 0; \tilde{\tau}) + \frac{1}{\tilde{\tau}}(2\nu + 1)\partial_{\tilde{\tau}} h(\nu, 0; \tilde{\tau}) + h(\nu, 0; \tilde{\tau}) = 0, \quad \tilde{\tau} = -k\tau \quad (3.3)$$

and $h(\nu, 1; -k\tau)$ is defined by $\partial_{\tilde{\tau}} h(\nu, 0; -k\tau)$. To cancel the $1/\tau^2$ term in the differential equation (2.5), one need multiply a prefactor $(-k\tau)^{\pm\nu}$ in the definition of h . Here we choose $(-k\tau)^{-\nu}$. Another key point of (3.3) is that h depends only on the combination of $k\tau$, not individual k and τ . As it will be seen, this construction extremely simplify the IBP relations of vertex integral family. We even find coincidentally that this definition automatically gives dlog-form DE of all vertex integral family. We will show the details later.

Due to the fact that u and u^* , or equivalently, $H_{\nu}^{(1)}(-k\tau)$ and $H_{\nu^*}^{(2)}(-k\tau)$ satisfy the same differential equation for the natural condition $k \in \mathbb{R}$ and $\nu = \pm\nu^*$ (or says ν are real or pure imaginary)², they also share the same properties under IBP. Thus we use the same symbol to denote both of them here for convenience. We call the integral family consists of these $V(\nu_0 + a_0, a_1, a_2, \dots, a_n)$ s for selected ν_i and k_i an **n-fold** (Hankel function) vertex integral family.

To apply IBP method, we need know the action of differential operators on $h(\nu, a, -k\tau)$. For ∂_{τ} , we have

$$\begin{aligned} \partial_{\tau} h(\nu, 0, -k\tau) &= -k h(\nu, 1, -k\tau), \\ \partial_{\tau} h(\nu, 1, -k\tau) &= -k \left[\frac{1}{k\tau} (2\nu + 1) h(\nu, 1; -k\tau) - h(\nu, 0; -k\tau) \right]. \end{aligned} \quad (3.4)$$

To construct IBP relation at loop-level corresponding to dl_i , and also construct DE with respect to k_i , we also need know the result of applying ∂_{k_i} to $h(\nu, a, -k\tau)$. One can use the properties of Hankel function to get the result, but a much simpler way is to notice that $h(\nu, a; \tilde{\tau})$'s dependence on k and τ are only mediated through $\tilde{\tau}$ as given in (3.3). Thus one immediately get

$$\begin{aligned} \partial_k h(\nu, 0, -k\tau) &= -\tau h(\nu, 1, -k\tau), \\ \partial_k h(\nu, 1, -k\tau) &= -\tau \left[\frac{1}{k\tau} (2\nu + 1) h(\nu, 1; -k\tau) - h(\nu, 0; -k\tau) \right]. \end{aligned} \quad (3.5)$$

With (3.4) and (3.5), IBP relations and DE corresponding to ∂_k could be easily constructed.

Now we want to show some important properties of this IBP system. Let us consider expressions involving only G_{+-} or/and G_{-+} type propagators. In this case, the correlator factorizes to

$$\int \prod_j \left[\hat{V}_j(\dots; \tau_j) d\tau_j \right] f(l_1, \dots, l_L) \prod_i dl_i \quad (3.6)$$

²One can see it easily by checking the first line of (2.5) where only ν^2 appears in the differential equation. With this condition in the definition of $h(\nu, 0; -k\tau)$, the prefactor $(-k\tau)^{-\nu}$ will be same for both $H_{\nu}^{(1)}(-k\tau)$ and $H_{\nu^*}^{(2)}$. Otherwise, the prefactor needs to be changed accordingly.

where f is polynomial of l_i and $\hat{V}$ is given in (3.1). The IBP relation with respect to a selected τ_k is given by

$$\int \left(\partial_{\tau_k} \hat{V}_k(\cdots; \tau_k) d\tau_k \right) \times \prod_{j \neq k} \left(\hat{V}_j(\cdots; \tau_j) d\tau_j \right) f(l_1, \cdots, l_L; k_1 \cdots, k_E) \prod_i dl_i \quad (3.7)$$

which does not affect other parts. We say the IBP relations with respect to different τ_i are factorized.

Now let's consider G_{++} and G_{--} type propagators. Its factorization is not as good as $G_{\pm\mp}$, since derivative acting on θ -function leads to the term

$$\int \hat{V}_i(\cdots; \tau_i) \delta(\tau_i - \tau_j) \hat{V}_j(\cdots; \tau_j) d\tau_i d\tau_j \times \cdots = \int \hat{V}_i(\cdots; \tau_i) \hat{V}_j(\cdots; \tau_i) d\tau_i \times \cdots, \quad (3.8)$$

where naively $(n_i + n_j)$ -fold Hankel function emerges. However, as we will show later, in fact, it becomes $(n_i + n_j - 2)$ -fold Hankel function or vanishes completely. Fortunately, the new term is belong a sub-sector with less vertex integral family, by which we means the IBP of the sub-sector will not involve the original higher sector. So we can still reduce each vertex integral family individually. And these remaining terms belong to sub-sector are regarded as known results in this part of reduce which have been reduced using the IBP-reduction of sub-sector.

Now we consider the merging happened in (3.8) by considering the symmetry in $G_{\pm\pm}$ more carefully. There are four cases could appear in the IBP system which involves massive $G_{\pm\pm}$:

$$\begin{aligned} & \left[h^{(a)}(\nu, 0, -k\tau_i) (\partial_{\tau_i} \theta_{ij}) h^{(3-a)}(\nu, 0, -k\tau_j) + h^{(3-a)}(\nu, 0, -k\tau_i) (\partial_{\tau_i} \theta_{ji}) h^{(a)}(\nu, 0, -k\tau_j) \right] \times \cdots \\ & \left[h^{(a)}(\nu, 1, -k\tau_i) (\partial_{\tau_i} \theta_{ij}) h^{(3-a)}(\nu, 0, -k\tau_j) + h^{(3-a)}(\nu, 1, -k\tau_i) (\partial_{\tau_i} \theta_{ji}) h^{(a)}(\nu, 0, -k\tau_j) \right] \times \cdots \\ & \left[h^{(a)}(\nu, 0, -k\tau_i) (\partial_{\tau_i} \theta_{ij}) h^{(3-a)}(\nu, 1, -k\tau_j) + h^{(3-a)}(\nu, 0, -k\tau_i) (\partial_{\tau_i} \theta_{ji}) h^{(a)}(\nu, 1, -k\tau_j) \right] \times \cdots \\ & \left[h^{(a)}(\nu, 1, -k\tau_i) (\partial_{\tau_i} \theta_{ij}) h^{(3-a)}(\nu, 1, -k\tau_j) + h^{(3-a)}(\nu, 1, -k\tau_i) (\partial_{\tau_i} \theta_{ji}) h^{(a)}(\nu, 1, -k\tau_j) \right] \times \cdots \\ & \theta_{ij} \equiv \theta(\tau_i - \tau_j), \quad a = 1, 2. \end{aligned} \quad (3.9)$$

The integration $\int d\tau_i d\tau_j$ of first and fourth cases will vanish due to

$$\begin{aligned} & \int d\tau_i d\tau_j \left[h^{(a)}(\nu, 0, -k\tau_i) h^{(3-a)}(\nu, 0, -k\tau_j) - h^{(3-a)}(\nu, 0, -k\tau_i) h^{(a)}(\nu, 0, -k\tau_j) \right] \delta(\tau_i - \tau_j) \times \cdots \\ & = \int d\tau_i \left[h^{(a)}(\nu, 0, -k\tau_i) h^{(3-a)}(\nu, 0, -k\tau_i) - h^{(3-a)}(\nu, 0, -k\tau_i) h^{(a)}(\nu, 0, -k\tau_i) \right] \times \cdots = 0 \\ & \int d\tau_i d\tau_j \left[h^{(a)}(\nu, 1, -k\tau_i) h^{(3-a)}(\nu, 1, -k\tau_j) - h^{(3-a)}(\nu, 1, -k\tau_i) h^{(a)}(\nu, 1, -k\tau_j) \right] \delta(\tau_i - \tau_j) \times \cdots \\ & = \int d\tau_i \left[h^{(a)}(\nu, 1, -k\tau_i) h^{(3-a)}(\nu, 1, -k\tau_i) - h^{(3-a)}(\nu, 1, -k\tau_i) h^{(a)}(\nu, 1, -k\tau_i) \right] \times \cdots = 0. \end{aligned} \quad (3.10)$$

The integration $\int d\tau_i d\tau_j$ of second and third cases give

$$\begin{aligned}
& -(-1)^a \int d\tau_i [F(-k\tau_i)] \times \dots = -(-1)^a \int d\tau_i e^{\pi \text{Im}[\nu] \frac{4i}{\pi}} (-k\tau_i)^{-2\nu-1} \times \dots \\
& +(-1)^a \int d\tau_i [F(-k\tau_i)] \times \dots = +(-1)^a \int d\tau_i e^{\pi \text{Im}[\nu] \frac{4i}{\pi}} (-k\tau_i)^{-2\nu-1} \times \dots \\
& F(-k\tau_i) = h^{(1)}(\nu, 1, -k\tau_i) h^{(2)}(\nu, 0, -k\tau_i) - h^{(2)}(\nu, 1, -k\tau_i) h^{(1)}(\nu, 0, -k\tau_i) \quad (3.11)
\end{aligned}$$

for real or imaginary ν , where we have used the property of Hankel function

$$F(-k\tau_i) = e^{\pi \text{Im}[\nu] \frac{4i}{\pi}} (-k\tau_i)^{-2\nu-1}. \quad (3.12)$$

One can also check it that the derivative of $F(-k\tau_i)$ gives $\partial_{-k\tau} F(-k\tau) = -\frac{2\nu+1}{-k\tau} F(-k\tau)$, then, $F(-k\tau_i) = C(-k\tau_i)^{-2\nu-1}$ and C can be determined by asymptotic behavior of Hankel functions on the boundary. Then, two vertex are "pinched" together and give a $(n_i + n_j - 2)$ -fold vertex integral family, not $(n_i + n_j)$ -fold from the naive observation in (3.8).

For massless $G_{\pm\pm}$, if one choose to keep the combination of the two terms in these propagator, there are also "vanishing" case and "pinched" case. "Vanishing" case comes from

$$\int d\tau_i d\tau_j \left[e^{ik\tau_i} (\partial_{\tau_i} \theta_{ij}) e^{-ik\tau_j} + e^{-ik\tau_i} (\partial_{\tau_i} \theta_{ji}) e^{ik\tau_j} \right] \times \dots = 0 \quad (3.13)$$

and "pinched" cases comes from

$$\begin{aligned}
& \int d\tau_i d\tau_j \left[(\partial_{\tau_i} e^{ik\tau_i}) (\partial_{\tau_i} \theta_{ij}) e^{-ik\tau_j} + (\partial_{\tau_i} e^{-ik\tau_i}) (\partial_{\tau_i} \theta_{ji}) e^{ik\tau_j} \right] \times \dots = \int d\tau_i 2ik \times \dots \\
& \int d\tau_i d\tau_j \left[e^{ik\tau_i} (\partial_{\tau_i} \theta_{ij}) \partial_{\tau_i} e^{-ik\tau_j} + e^{-ik\tau_i} (\partial_{\tau_i} \theta_{ji}) \partial_{\tau_i} e^{ik\tau_j} \right] \times \dots = - \int d\tau_i 2ik \times \dots \quad (3.14)
\end{aligned}$$

One can also choose to handle the two terms in massless propagator individually.

With the above discussion, we can reduce each vertex integral family individually:

$$\int \prod_i d\tau_i \hat{V}_i \times \dots = \left[\int \prod_i \left(d\tau_i \sum_j c_j^{(i)} \hat{f}_j^{(i)} \right) \times \dots \right] + R, \quad (3.15)$$

where $\hat{f}_j^{(i)}$ is the integrand of the MI of the integral family V_i belong to, "..." part could contain $\theta(\tau_i - \tau_j)$, and we use R to denote remaining terms. R terms come from pinching two vertex connected via $G_{\pm\pm}$ together. Since the reduction of these R terms will not involve the family before pinching, it belongs to a sub-sector and can treat as known in the reduction of family V_i in (3.15). This form leads following simplifications in the computation:

- **Factorization of IBP relations of vertex:** We say that the IBP relations of different $d\tau_i$ are factorized in the $(t, \mathbf{k})$ space of dS integral, since we can perform the IBP reduction individually for each of them, as shown in (3.15). Since all vertex

integral family take the similar expression (3.1), the reduction result can be used repeatedly for different vertex, correlator and theory. For example, once people get the IBP reduction of a n-fold vertex integral family, it can be directly applied to all n-fold vertex appears in other place (if without external symmetry).

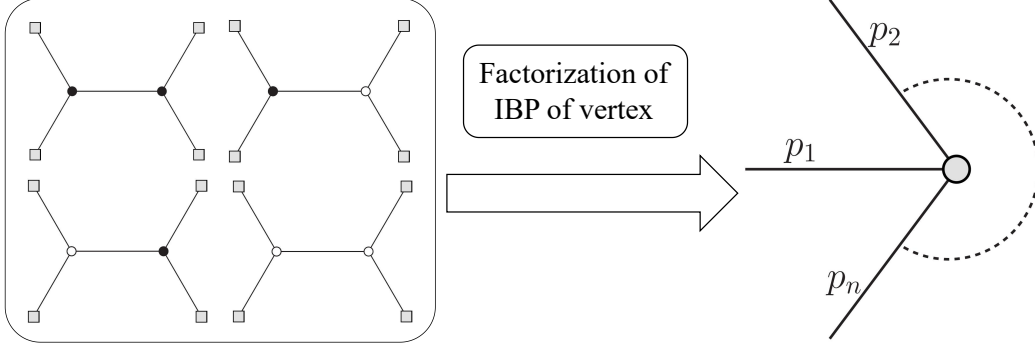


Figure 1. Factorization of IBP relations of vertex: all reductions with respect to $d\tau$ can be achieved by reducing individual families of vertex integrals separately.

- At loop-level, one need to further consider the IBP relations of dl_i . Notice that terms outside of $\hat{V}_i$ and θ -function are independent of τ_i . Due to this, one can reduce the family of all $d\tau$ integrals to MI first, to get $d\tau_i$ -reduced IBP relations. Then, solving these $d\tau_i$ -reduced IBP relations of dl_i will give the complete result of reduction. In

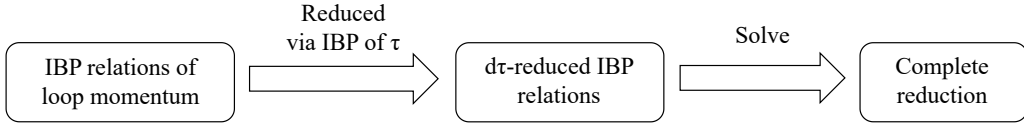


Figure 2. Steps of loop-level reduction

this progress, differential equations of $d\tau$ integrals family could serve for directly giving $d\tau_i$ -reduced IBP relations. For example, consider total derivative with respect to l for the following loop integrals

$$\int d \left(\frac{1}{|l||l-p_1|\dots} \mathbf{f} \right) = 0, \quad \mathbf{f} = \{f_1, f_2, \dots\} \quad (3.16)$$

where the $\mathbf{f}$ are MI of the integrals family of all $d\tau_i$. Then, we have a series of $d\tau_i$ -reduced IBP relations

$$\int \left(d \frac{1}{|l||l-p_1|\dots} \right) \mathbf{f} + \frac{1}{|l||l-p_1|\dots} (\Omega \cdot \mathbf{f} dl) = 0, \quad d\mathbf{f} = \Omega \cdot \mathbf{f} dl \quad (3.17)$$

where the Ω is the differential equation matrix of $\mathbf{f}$ with respect to l . (3.17) is directly the so called $d\tau$ -reduced IBP relation, because the $d\tau$ integration part of each integral

in these relations are kept as master integrals.

In the rest of this section, we will show some examples for n-fold vertex integral family and sketch more features of this IBP system. Following the idea similar to iterative reduction [79, 80] in flat space, we will further see that the infinite number of integrals in a vertex integral family can be reduced iteratively to MI once we have solved the iterative relations from a finite linear system. Furthermore, we get the universal expression of these iterative reductions for any n-fold vertex integral family, as well as d log-form DE of MI. We also will give formulas and discussion for reduction with remaining terms come from $G_{\pm\pm}$.

3.2 1-fold vertex integral family: iterative reduction

Consider 1-fold vertex integral family

$$V(\nu_0, a_1) = \int_{-\infty}^0 \tau^{\nu_0} e^{ik_0\tau} h(\nu_1, a_1; -k_1\tau) d\tau, \quad a_i = 0, 1, \quad (3.18)$$

From now on we set H to be 1 for convenience. We use $eq[\nu_0, \mathbf{a}]$ to denote the IBP relation

$$\int d\hat{V}(\nu_0, \mathbf{a}; \tau) = 0, \quad (3.19)$$

then we have IBP relations

$$\begin{aligned} eq[\nu_0, 0] : ik_0 V(\nu_0, 0) + \nu_0 V(\nu_0 - 1, 0) - k_1 V(\nu_0, 1) &= 0 \\ eq[\nu_0, 1] : ik_0 V(\nu_0, 1) + \nu_0 V(\nu_0 - 1, 1) + k_1 V(\nu_0, 0) + (-2\nu_1 - 1) V(\nu_0 - 1, 1) &= 0. \end{aligned} \quad (3.20)$$

We have numerically evaluated these integrals at their convergent region and verified these two IBP relations to be right. Defining

$$\mathbf{f}^{(a_0)} = \{V(\nu_0 + a_0, 0), V(\nu_0 + a_0, 1)\}, \quad (3.21)$$

the IBP relation corresponding to a selected ν_0 can be expressed in the form of matrix as

$$\begin{aligned} (M_1^{(1)} + \nu_0 \mathbb{I}_2) \cdot \mathbf{f}^{(-1)} + (M_0^{(1)} + ik_0 \mathbb{I}_2) \cdot \mathbf{f}^{(0)} &= 0 \\ M_1^{(j)} = -\frac{2\nu_j + 1}{2} (\mathbb{I}_2 - \sigma_3) = \begin{pmatrix} 0 & 0 \\ 0 & -2\nu_j - 1 \end{pmatrix}, \quad M_0^{(j)} = -ik_j \sigma_2 = \begin{pmatrix} 0 & -k_j \\ k_j & 0 \end{pmatrix}, \end{aligned} \quad (3.22)$$

where $\sigma_{1,2,3}$ are the Pauli matrices. Traditionally, taking ν_0 to be $\nu_0 + a_0$ with different a_0 , we can get a series of IBP relations. Solving them give the reduction of this integral family. As the consequence, we find there are two MI in this system, which can be chosen as $\mathbf{f}^{(0)}$. However, here we obviously have a simpler method. People only need to solve iterative relation

$$\mathbf{f}^{(-1)} = A_-(\nu_0) \cdot \mathbf{f}^{(0)}, \quad (3.23)$$

Then, all integrals in the family can be reduced iteratively by

$$\begin{aligned} \mathbf{f}^{(-n)} &= \left(\prod_{i=n-1}^0 A_-(\nu_0 - i) \right) \cdot \mathbf{f}^{(0)}, \\ \mathbf{f}^{(n)} &= \left(\prod_{i=n-1}^0 A_+(\nu_0 + i) \right) \cdot \mathbf{f}^{(0)}, \quad A_+(\nu_0) \equiv (A_-(\nu_0 + 1))^{-1}, \end{aligned} \quad (3.24)$$

For 1-fold vertex integral family, $A_{\pm}$ can be immediately solved and given as following

$$\begin{aligned} A_-(\nu_0) &= \begin{pmatrix} -\frac{ik_0}{\nu_0} & \frac{k_1}{\nu_0} \\ \frac{k_1}{-\nu_0+2\nu_1+1} & -\frac{ik_0}{\nu_0-2\nu_1-1} \end{pmatrix} \\ A_+(\nu_0) &= \begin{pmatrix} \frac{ik_0(\nu_0+1)}{k_0^2-k_1^2} & \frac{k_1(\nu_0-2\nu_1)}{k_0^2-k_1^2} \\ \frac{k_1(\nu_0+1)}{k_1^2-k_0^2} & \frac{ik_0(\nu_0-2\nu_1)}{k_0^2-k_1^2} \end{pmatrix} \end{aligned} \quad (3.25)$$

For $k_0 = 0$, they will be simplified to

$$A_-(\nu_0) = \begin{pmatrix} 0 & \frac{k_1}{\nu_0} \\ \frac{k_1}{-\nu_0+2\nu_1+1} & 0 \end{pmatrix}, \quad A_+(\nu_0) = \begin{pmatrix} 0 & -\frac{\nu_0-2\nu_1}{k_1} \\ \frac{\nu_0+1}{k_1} & 0 \end{pmatrix}, \quad (3.26)$$

3.3 1-fold vertex integral family: d log-form DE

Using above reduction we can establish the DE for the MI by acting ∂_{k_0} and ∂_{k_1} on $\mathbf{f}^{(0)}$. They gives

$$\begin{aligned} \partial_{k_0} \mathbf{f}^{(0)} &= \begin{pmatrix} iV(\nu_0+1, 0) \\ iV(\nu_0+1, 1) \end{pmatrix} = iA_+(\nu_0+1) \mathbf{f}^{(0)}, \\ \partial_{k_1} \mathbf{f}^{(0)} &= \begin{pmatrix} -V(\nu_0+1, 1) \\ \frac{(-2\nu_1-1)V(\nu_0, 1)}{k_1} + V(\nu_0+1, 0) \end{pmatrix} = \left(\frac{1}{k_1} M_1^{(1)} - i\sigma_2 \cdot A_+(\nu_0+1) \right) \cdot \mathbf{f}^{(0)}. \end{aligned} \quad (3.27)$$

Evidently, the partial derivatives do not give rise to any special functions beyond the original family of functions, this implies that DE method corresponding to ∂_{k_i} can be applied for such integral family. Let's reduce the right hand side of these equations with the iterative relation we get in (3.25). It gives

$$\begin{aligned} \partial_{k_i} \mathbf{f}^{(0)} &= \Omega_{k_i} \cdot \mathbf{f}^{(0)}, \\ \Omega_{k_0} &= \begin{pmatrix} -\frac{k_0(\nu_0+1)}{k_0^2-k_1^2} & \frac{ik_1(\nu_0-2\nu_1)}{k_0^2-k_1^2} \\ -\frac{ik_1(\nu_0+1)}{k_0^2-k_1^2} & -\frac{k_0(\nu_0-2\nu_1)}{k_0^2-k_1^2} \end{pmatrix}, \\ \Omega_{k_1} &= \begin{pmatrix} \frac{k_1(\nu_0+1)}{k_0^2-k_1^2} & -\frac{ik_0(\nu_0-2\nu_1)}{k_0^2-k_1^2} \\ -\frac{ik_0k_1(\nu_0+1)}{k_1^3-k_0^2k_1} & \frac{k_0^2(2\nu_1+1)-k_1^2(\nu_0+1)}{k_1^3-k_0^2k_1} \end{pmatrix}. \end{aligned} \quad (3.28)$$

In flat space, canonical DE play an important role in analytical calculation of loop integral. It takes the form

$$d\mathbf{f} = \varepsilon(d\Omega) \cdot \mathbf{f} \quad (3.29)$$

with $d\Omega$ in $d\log$ -form. It is interesting that we find in 1-fold vertex integral family, in the MI we defined, the DE automatically is in $d\log$ -form. Although it is subtly different from canonical DE, we report this elegant result as it may offer potential assistance in understanding the mathematical structure of dS integrals in the future. The DE can be written in $d\log$ -form as

$$\begin{aligned} d\mathbf{f}^{(0)} &= d\Omega \mathbf{f}^{(0)} \\ d\Omega &= \sum_{i=0,1} \tilde{\Omega}_{k_i} dk_i = C_1 d\log(k_1) + C_2 d\log[(k_0 - k_1)(k_0 + k_1)] + C_3 d\log\left(\frac{k_0 + k_1}{k_0 - k_1}\right) \\ C_1 &= \begin{pmatrix} 0 & 0 \\ 0 & -2\nu_1 - 1 \end{pmatrix}, \quad C_2 = \begin{pmatrix} \frac{1}{2}(-\nu_0 - 1) & 0 \\ 0 & \frac{1}{2}(2\nu_1 - \nu_0) \end{pmatrix} \\ C_3 &= \begin{pmatrix} 0 & -\frac{1}{2}i(\nu_0 - 2\nu_1) \\ \frac{1}{2}i(\nu_0 + 1) & 0 \end{pmatrix}. \end{aligned} \quad (3.30)$$

where $df \equiv \sum_i \partial_{k_i} f dk_i$, and $d\nu_i$ is not included (i.e., we treat ν_i as fixed parameters).

3.4 n-fold vertex integral family: universal formula of iterative reduction

Now we consider n-fold vertex integral family

$$V(\nu_0, a_1, \dots, a_n) = \int_{-\infty}^0 \tau^{\nu_0} e^{ik_0\tau} \prod_{i=1}^n h(\nu_i, a_i; -k_i\tau) d\tau, \quad a_i = 0, 1. \quad (3.31)$$

It has 2^n MI, which can be chosen as

$$f_{\mathbf{a}}^{(0)} = V(\nu_0, \mathbf{a}), \quad \mathbf{a} = a_1, \dots, a_n, \quad \forall a_i = 0, 1. \quad (3.32)$$

and together denoted as $\mathbf{f}^{(0)}$. As a vector with 2^n components, the ordering is given by n indices a_i according to the natural binary number j

$$j = 1 + \sum_i a_i 2^{n-i}. \quad (3.33)$$

For example, when $n = 2$, we have

$$f_1^{(0)} = f_{0,0}^{(0)}, \quad f_2^{(0)} = f_{0,1}^{(0)}, \quad f_3^{(0)} = f_{1,0}^{(0)}, \quad f_4^{(0)} = f_{1,1}^{(0)}. \quad (3.34)$$

It can be derived from (3.4) that all IBP relations $eq[\nu_0, \mathbf{a}]$ corresponding to a selected

ν_0 can be expressed in the form

$$(M_1)_{\mathbf{ba}} f_{\mathbf{a}}^{(-1)} + (M_0)_{\mathbf{ba}} f_{\mathbf{a}}^{(0)} = 0 \quad (3.35)$$

where the matrix elements are given by

$$\begin{aligned} (M_1)_{\mathbf{ba}} &= \sum_{j=1}^n \left[\left(M_1^{(j)} \right)_{b_j a_j} \prod_{i \neq j} \delta_{b_i a_i} \right] + \nu_0 \delta_{\mathbf{ba}} \\ (M_0)_{\mathbf{ba}} &= \sum_{j=1}^n \left[\left(M_0^{(j)} \right)_{b_j a_j} \prod_{i \neq j} \delta_{b_i a_i} \right] + i k_0 \delta_{\mathbf{ba}} \end{aligned} \quad (3.36)$$

with $M_0^{(j)}, M_1^{(j)}$ given in (3.22). The matrix can be compactly represented as

$$\begin{aligned} M_1 &= \sum_{j=1}^n \left(\nu_j + \frac{1}{2} \right) \Lambda_3^{(j)} + \left(\nu_0 - \frac{n}{2} - \sum_{i=1}^n \nu_i \right) \mathbb{I}_{2^n} \\ M_0 &= -i \sum_{j=1}^n k_j \Lambda_2^{(j)} + i k_0 \mathbb{I}_{2^n} \end{aligned} \quad (3.37)$$

where

$$\left(\Lambda_k^{(j)} \right)_{\mathbf{ba}} \equiv (\sigma_k)_{b_j, a_j} \prod_{i \neq j} \delta_{b_i, a_i}, \quad k = 1, 2, 3 \quad (3.38)$$

is direct product of a series 2×2 identity matrices except the j -th one as Pauli σ_k matrix.

From the representation (3.37), one can see M_1 is a diagonal matrix

$$(M_1)_{\mathbf{ba}} = \begin{cases} \nu_0 - \sum_i a_i (2\nu_i + 1), & \mathbf{b} = \mathbf{a} \\ 0, & \mathbf{b} \neq \mathbf{a} \end{cases} \quad (3.39)$$

Here we show IBP relation $eq[\nu_0, \mathbf{a}]$ for $n = 2$ and all $\mathbf{a}$ in matrix form as example

$$\begin{aligned} & \left(M_1 \middle| M_0 \right) \cdot \mathbf{f}^\top \\ &= \begin{pmatrix} \nu_0 & 0 & 0 & 0 & \left| \begin{matrix} i k_0 & -k_2 & -k_1 & 0 \end{matrix} \right. \\ 0 & \nu_0 - 2\nu_2 - 1 & 0 & 0 & \left| \begin{matrix} k_2 & i k_0 & 0 & -k_1 \end{matrix} \right. \\ 0 & 0 & \nu_0 - 2\nu_1 - 1 & 0 & \left| \begin{matrix} k_1 & 0 & i k_0 & -k_2 \end{matrix} \right. \\ 0 & 0 & 0 & \nu_0 - 2\nu_1 - 2\nu_2 - 2 & \left| \begin{matrix} 0 & k_1 & k_2 & i k_0 \end{matrix} \right. \end{pmatrix} \cdot \mathbf{f}^\top = 0, \\ & \mathbf{f} = \{\mathbf{f}^{(-1)}, \mathbf{f}^{(0)}\}, \quad \mathbf{f}^{(i)} = \{f_{0,0}^{(i)}, f_{0,1}^{(i)}, f_{1,0}^{(i)}, f_{1,1}^{(i)}\}, \end{aligned} \quad (3.40)$$

where

$$M_1 = M_1^{(1)} \otimes \mathbb{I}_2 + \mathbb{I}_2 \otimes M_1^{(2)} + \nu_0 \mathbb{I}_2 \otimes \mathbb{I}_2$$

$$M_0 = M_0^{(1)} \otimes \mathbb{I}_2 + \mathbb{I}_2 \otimes M_0^{(2)} + ik_0 \mathbb{I}_2 \otimes \mathbb{I}_2. \quad (3.41)$$

Obviously,

$$A_-(\nu_0) = -M_1^{-1}.M_0, \quad A_+(\nu_0 - 1) = -M_0^{-1}.M_1. \quad (3.42)$$

In above expression, the inverse of diagonal matrix M_1 is trivial, but the inverse of M_0 is non-trivial. Thus we are looking for different representation. Noticing that M_0 is formed by the direct product of $\mathbb{I}_2$ and σ_2 , diagonalizing σ_2 by redefining each $h(\nu_i, a_i; -k\tau)$ immediately leads to the diagonalization of M_0 :

$$\tilde{h}(\nu_i, a_i; -k\tau) = \sum_{b_i=0,1} T_{a_i b_i} h(\nu_i, b_i; -k\tau) \quad (3.43)$$

with

$$T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}, \quad T^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad (3.44)$$

This leads to

$$\begin{aligned} \tilde{V}(\nu_0, \mathbf{a}) &= \int_{-\infty}^0 \tau^{\nu_0} e^{ik_0 \tau} \prod_i \tilde{h}(\nu_i, a_i; -k_i \tau) d\tau \\ \tilde{\mathbf{f}}^{(a_0)} &= T_n \cdot \mathbf{f}^{(a_0)}, \quad (T_n)_{ba} = \prod_{i=1}^n T_{b_i a_i}, \quad (T_n^{-1})_{ba} = \prod_{i=1}^n T_{b_i a_i}^{-1} \end{aligned} \quad (3.45)$$

Using the properties

$$\begin{aligned} T \cdot \sigma_2 \cdot T^{-1} &= \sigma_3, \quad T \cdot \sigma_3 \cdot T^{-1} = -\sigma_2, \quad T \cdot \sigma_1 \cdot T^{-1} = \sigma_1, \quad T \cdot \mathbb{I}_2 \cdot T^{-1} = \mathbb{I}_2, \\ T_n \cdot \Lambda_2^{(j)} \cdot T_n^{-1} &= \Lambda_3^{(j)}, \quad T_n \cdot \Lambda_3^{(j)} \cdot T_n^{-1} = -\Lambda_2^{(j)}. \end{aligned} \quad (3.46)$$

after applying the transformation (3.44) and (3.45) to (3.22), we have

$$\begin{aligned} \tilde{M}_1^{(j)} &= -\frac{2\nu_i + 1}{2} (\mathbb{I}_2 + \sigma_2), \quad \tilde{M}_0^{(j)} = -ik_j \sigma_3, \\ \tilde{M}_1 &= -\sum_{j=1}^n \left(\nu_j + \frac{1}{2} \right) \Lambda_2^{(j)} + \left(\nu_0 - \frac{n}{2} - \sum_{i=1}^n \nu_i \right) \mathbb{I}_{2^n}, \\ \tilde{M}_0 &= -\sum_{j=1}^n ik_j \Lambda_3^{(j)} + ik_0 \mathbb{I}_{2^n}. \end{aligned} \quad (3.47)$$

Now, on the contrary, $\tilde{M}_0$ is a diagonal matrix

$$\left(\tilde{M}_0 \right)_{ba} = \begin{cases} i(k_0 + \sum_{i=1}^n (2a_i - 1)k_i), & \mathbf{b} = \mathbf{a} \\ 0, & \mathbf{b} \neq \mathbf{a} \end{cases} \quad (3.48)$$

while $\tilde{M}_1$ is not. For example, when $n = 2$, we have

$$\begin{aligned} (\tilde{M}_1 | \tilde{M}_0) \cdot \tilde{\mathbf{f}}^\top &= 0, \quad \tilde{\mathbf{f}} = \{\tilde{\mathbf{f}}^{(-1)}, \tilde{\mathbf{f}}^{(0)}\}, \\ \tilde{M}_1 &= \begin{pmatrix} \nu_0 - \nu_1 - \nu_2 - 1 & \frac{1}{2}i(2\nu_2 + 1) & \frac{1}{2}i(2\nu_1 + 1) & 0 \\ -\frac{1}{2}i(2\nu_2 + 1) & \nu_0 - \nu_1 - \nu_2 - 1 & 0 & \frac{1}{2}i(2\nu_1 + 1) \\ -\frac{1}{2}i(2\nu_1 + 1) & 0 & \nu_0 - \nu_1 - \nu_2 - 1 & \frac{1}{2}i(2\nu_2 + 1) \\ 0 & -\frac{1}{2}i(2\nu_1 + 1) & -\frac{1}{2}i(2\nu_2 + 1) & \nu_0 - \nu_1 - \nu_2 - 1 \end{pmatrix} \\ \tilde{M}_0 &= i \begin{pmatrix} k_0 - k_1 - k_2 & 0 & 0 & 0 \\ 0 & k_0 - k_1 + k_2 & 0 & 0 \\ 0 & 0 & k_0 + k_1 - k_2 & 0 \\ 0 & 0 & 0 & k_0 + k_1 + k_2 \end{pmatrix} \end{aligned} \quad (3.49)$$

Putting all together finally we have

$$\begin{aligned} A_-(\nu_0) &= -M_1^{-1} \cdot M_0, \\ A_+(\nu_0 - 1) &= -T_n^{-1} \cdot \tilde{M}_0^{-1} \cdot \tilde{M}_1 \cdot T_n = -T_n^{-1} \cdot \tilde{M}_0^{-1} \cdot T_n \cdot M_1, \end{aligned} \quad (3.50)$$

where M_1^{-1} , $\tilde{M}_0^{-1}$ and their inverse are diagonal matrices, with M_1 merely relying on ν_i , $\tilde{M}_0$ only relying on k_i and T_n a constant matrix.

Before ending this subsection, let us give a remark. Let's see what will happen if we haven't asked the definition of $h(\mu, \mathbf{a})$ to cancel the $1/\tau^2$ in (3.2). For example, define $h(\nu_0, \mathbf{a}) = (-k\tau)^{\frac{1}{2}} H_\nu^{(1,2)}$ in (3.1) which satisfy

$$\partial_{\tilde{\tau}}^2 h(\mu, 0; \tilde{\tau}) + \left(1 + \frac{\mu^2}{\tilde{\tau}^2}\right) h(\mu, 0; \tilde{\tau}) = 0, \quad \tilde{\tau} = -k\tau, \quad \mu^2 = \frac{m^2}{H^2} - 2 \quad (3.51)$$

we will have

$$\begin{aligned} M_2^{(1)} \cdot \mathbf{f}^{(-2)} + \nu_0 \mathbb{I}_2 \cdot \mathbf{f}^{(-1)} + (M_0^{(1)} + ik_0 \mathbb{I}_2) \cdot \mathbf{f}^{(0)} \\ M_2^{(j)} = \begin{pmatrix} 0 & 0 \\ \frac{\mu_j^2}{k_j} & 0 \end{pmatrix}, \quad M_0^{(j)} = \begin{pmatrix} 0 & -k_j \\ k_j & 0 \end{pmatrix}. \end{aligned} \quad (3.52)$$

The formula of reduction will not be as explicit as it is now due to the emergence of $\mathbf{f}^{(-2)}$ from the $1/\tau^2$ term. That's why we request this term to be canceled in the definition.

3.5 n-fold vertex integral family: universal formula of d log-form DE

For n-fold vertex integral family, with (3.5), acting ∂_{k_0} and ∂_{k_i} on $\mathbf{f}^{(0)}$ give

$$\begin{aligned} \partial_{k_0} \mathbf{f}^{(0)} &= i \mathbf{f}^{(1)} = i A_+(\nu_0) \mathbf{f}^{(0)}, \\ \partial_{k_i} \mathbf{f}^{(0)} &= \left(-\frac{1}{k_i} \frac{2\nu_i + 1}{2} (\mathbb{I}_{2^n} - \Lambda_3^{(i)}) - i \Lambda_2^{(i)} \cdot A_+(\nu_0) \right) \cdot \mathbf{f}^{(0)}, \quad \text{for } i > 0. \end{aligned} \quad (3.53)$$

Defining

$$\begin{aligned} (\tilde{\Omega}_0)_{ba} &\equiv \begin{cases} -i \log [k_0 + \sum_i (2a_i - 1)k_i], & \mathbf{b} = \mathbf{a} \\ 0, & \mathbf{b} \neq \mathbf{a} \end{cases}, \\ (\Omega_{ex})_{ba} &\equiv \begin{cases} -\sum_i a_i (2\nu_i + 1) \log k_i, & \mathbf{b} = \mathbf{a} \\ 0, & \mathbf{b} \neq \mathbf{a} \end{cases}, \end{aligned} \quad (3.54)$$

we have d log-form DE

$$\begin{aligned} d\mathbf{f}^{(0)} &= (d\Omega) \cdot \mathbf{f}^{(0)} = \sum_{i=0}^n \Omega_{k_i} \cdot \mathbf{f}^{(0)} dk_i, \\ \Omega &= \Omega_{ex} - iT_n^{-1} \cdot \tilde{\Omega}_0 \cdot T_n \cdot M_1(\nu_0 + 1), \end{aligned} \quad (3.55)$$

where $M_1(\nu_0 + 1)$ is shifting the ν_0 in the original M_1 to $\nu_0 + 1$.

Proof:

$$\begin{aligned} \partial_{k_0} \Omega &= -iT_n^{-1} \cdot \partial_{k_0} \tilde{\Omega}_0 \cdot T_n \cdot M_1(\nu_0 + 1) = -iT_n^{-1} \cdot \tilde{M}_0^{-1} \cdot T_n \cdot M_1(\nu_0 + 1) = iA_+(\nu_0) \\ \partial_{k_i} \Omega &= \partial_{k_i} \Omega_{ex} - iT_n^{-1} \cdot \partial_{k_i} \tilde{\Omega}_0 \cdot T_n \cdot M_1(\nu_0 + 1) \\ &= -\frac{1}{k_i} \frac{2\nu_i + 1}{2} (\mathbb{I}_{2^n} - \Lambda_3^{(i)}) + iT_n^{-1} \cdot \Lambda_3^{(i)} \cdot \tilde{\Omega}_0 \cdot T_n \cdot M_1(\nu_0 + 1) \\ &= -\frac{1}{k_i} \frac{2\nu_i + 1}{2} (\mathbb{I}_{2^n} - \Lambda_3^{(i)}) + i\Lambda_2^{(i)} \cdot T_n^{-1} \cdot \tilde{\Omega}_0 \cdot T_n \cdot M_1(\nu_0 + 1) \\ &= -\frac{1}{k_i} \frac{2\nu_i + 1}{2} (\mathbb{I}_{2^n} - \Lambda_3^{(i)}) - i\Lambda_2^{(i)} \cdot A_+(\nu_0). \end{aligned} \quad (3.56)$$

Comparing these equations with (3.53), the proof is complete.

Then, once people have got boundary condition of $\mathbf{f}^{(0)}$, $\mathbf{f}^{(0)}(k_0^0, k_1^0, \dots)$ for example, $\mathbf{f}^{(0)}(k'_0, k'_1, \dots)$ can be got by

$$\begin{aligned} \mathbf{f}^{(0)}(k'_0, k'_1, \dots) &= \mathbf{f}^{(0)}(k_0^0, k_1^0, \dots) \\ &+ \mathcal{P} \exp \left[\int_{(k_0^0, k_1^0, \dots)}^{(k'_0, k'_1, \dots)} \sum_i \Omega_{k_i}(\nu_0, \nu_1, \dots; k_0, k_1, \dots) dk_i \right] \cdot \mathbf{f}^{(0)}(k_0^0, k_1^0, \dots) \end{aligned} \quad (3.57)$$

for a integration path starts from $(k_1^0, k_2^0, \dots)$ and end at $(k'_1, k'_2, \dots)$. $\mathcal{P}$ for path ordering.

At loop-level, these differential equations with respect to loop momentum will also directly give the $d\tau_i$ -reduced IBP relations we have mentioned near the end of section 3.1.

3.6 0-fold vertex integral family: massless cases

For a vertex only connect to massless propagator, the vertex function family is given by

$$V(\nu_0) = \int \tau^{\nu_0} e^{ik_0 \tau}. \quad (3.58)$$

It has only $2^0 = 1$ MI and the iterative reduction relations are given by

$$\begin{aligned} eq[\nu_0] : ik_0 V(\nu_0) + \nu_0 V(\nu_0 - 1) &= 0, \\ A_+(\nu_0) &= i \frac{\nu_0 + 1}{k_0}, \quad A_-(\nu_0) = -i \frac{k_0}{\nu_0}. \end{aligned} \quad (3.59)$$

and d log-form DE is given by

$$\begin{aligned} \partial_{k_0} V(\nu_0) &= iV(\nu_0 + 1) = -(\nu_0 + 1) \frac{1}{k_0} V(\nu_0), \\ d\Omega &= \Omega_{k_0} dk_0 = -(\nu_0 + 1) d \log(k_0). \end{aligned} \quad (3.60)$$

3.7 Discussion on remaining terms come from $G_{\pm\pm}$

3.7.1 Iterative reduction

Consider an simple example with remaining terms. Suppose $f_{\mathbf{a};1}^{(a_0)}$ are integrals belong to vertex integral family V_1 , corresponding to the integration of $d\tau_1$, and $f_{\mathbf{b};2}^{(b_0)}$ are integrals belong to vertex integral family V_2 . We use $\hat{f}$ to denote their integrand. For example

$$\hat{f}_{\mathbf{a};1}^{(a_0)} = e^{ik_{0;1}\tau_1} \tau_1^{\nu_{0;1}+a_0} \prod_i h(\nu_{i;1}, a_i, -k_{i;1}\tau_1). \quad (3.61)$$

We also denote the combination of two terms in propagator as

$$\begin{aligned} h(\nu_{i;1}, a_i, -k_{i;1}\tau_1) \theta_{1,2}^{(i,j)} h(\nu_{j;2}, b_j, -k_{j;2}\tau_2) &\equiv \\ h^{(1)}(\nu_{i;1}, a_i, -k_{i;1}\tau_1) \theta_{12} h^{(2)}(\nu_{j;2}, b_j, -k_{j;2}\tau_2) &+ h^{(2)}(\nu_{i;1}, a_i, -k_{i;1}\tau_1) \theta_{21} h^{(1)}(\nu_{j;2}, b_j, -k_{j;2}\tau_2), \\ \nu_{i;1} = \nu_{j;2}, \quad k_{i;1} = k_{j;2}. \end{aligned} \quad (3.62)$$

Then, if there is a G_{++} , we denote the integrals as

$$\mathbf{f}_{\mathbf{a},\mathbf{b}}^{(a_0,b_0)} \equiv \int d\tau_1 d\tau_2 \hat{f}_{\mathbf{a};1}^{(a_0)} \theta_{1,2}^{(i,j)} \hat{f}_{\mathbf{b};2}^{(b_0)} \quad (3.63)$$

Without loss of generality, consider IBP of $d\tau_1$ and $a_0 = b_0 = 0$, one could have IBP relations

$$(M_1)_{ca} \mathbf{f}_{\mathbf{a},\mathbf{b}}^{(-1,0)} + (M_0)_{ca} \mathbf{f}_{\mathbf{a},\mathbf{b}}^{(0,0)} + \delta_{ca} \mathbf{R}_{\mathbf{a},\mathbf{b}}^{(0,0)} = 0, \quad (3.64)$$

which is just adding remaining terms of sub-sector and the factor $\theta_{1,2}^{(i,j)} \hat{f}_{\mathbf{b};2}^{(0)}$ to (3.35). According to (3.10) and (3.11),

$$\begin{aligned} \mathbf{R}_{\mathbf{a},\mathbf{b}}^{(a_0,b_0)} &= \delta_{a_i,1-b_j} (-1)^{a_i+1} \frac{4i}{\pi} e^{\pi \text{Im}[\nu]} (-k_{i;1})^{-2\nu_{i;1}-1} f_{\mathbf{a}_{\hat{i}}, \mathbf{b}_{\hat{j}}}^{(a_0+b_0-2\nu_{i;1}-1)}, \\ \mathbf{a}_{\hat{i}}, \mathbf{b}_{\hat{j}} &= a_1, a_2, \dots, a_{i-1}, a_{i+1}, \dots, a_{n_1}, b_1, \dots, b_{j-1}, b_{j+1}, \dots, b_{n_2}, \end{aligned} \quad (3.65)$$

where $f_{\mathbf{a}_i, \mathbf{b}_j}^{(a_0+b_0-2\nu_{i;1}-1)}$ is integral belong to $(n_1 + n_2 - 2)$ -fold vertex integral family. Then, reduction of these sector with two vertex integral family could be achieved via using

$$\begin{aligned} \mathbf{f}_{\mathbf{c}, \mathbf{b}}^{(-1,0)} &= (A_{-;1}(\nu_0))_{\mathbf{ca}} \mathbf{f}_{\mathbf{a}, \mathbf{b}}^{(0,0)} - \left(M_{1;1}^{-1}\right)_{\mathbf{ca}} \mathbf{R}_{\mathbf{a}, \mathbf{b}}^{(0,0)}, \\ \mathbf{f}_{\mathbf{c}, \mathbf{b}}^{(1,0)} &= (A_{+;1}(\nu_0))_{\mathbf{ca}} \mathbf{f}_{\mathbf{a}, \mathbf{b}}^{(0,0)} - \left(T_{n_1}^{-1} \cdot \tilde{M}_{0;1}^{-1} \cdot T_{n_1}\right)_{\mathbf{ca}} \mathbf{R}_{\mathbf{a}, \mathbf{b}}^{(1,0)} \end{aligned} \quad (3.66)$$

iteratively. In these equations, the subscript ;1 means they are the expression with respect to family V_1 . In other words, the $A_{\pm;1}$, $M_{1;1}$, and $\tilde{M}_{0;1}$ here are just given by the same formulas as (3.50), (3.37), and (3.47). These formulas give the iterative reduction relations without computing inverse of large matrix. For more than one $G_{\pm\pm}$, the reduction is similar to (3.66), but with more remaining terms corresponding to each $G_{\pm\pm}$. Since all remaining terms are also product of vertex integral families, one could reduce them via formula like (3.66) as well, thus the reduction is complete. We have got the universal formula for reducing all tree-level dS integrals.

3.7.2 d log-form DE

Consider the example (3.63) again, DE is given by

$$\begin{aligned} \left(\partial_{k_{0;1}} \mathbf{f}^{(0,0)}\right)_{\mathbf{c}, \mathbf{b}} &= i \mathbf{f}_{\mathbf{c}, \mathbf{b}}^{(1,0)}, \\ \left(\partial_{k_{i;1}} \mathbf{f}^{(0,0)}\right)_{\mathbf{c}, \mathbf{b}} &= -\frac{1}{k_i} \frac{2\nu_i + 1}{2} \left(\mathbb{I}_{2^n} - \Lambda_3^{(i)}\right)_{\mathbf{ca}} \mathbf{f}_{\mathbf{a}, \mathbf{b}}^{(0,0)} - i \left(\Lambda_2^{(i)}\right)_{\mathbf{ca}} \mathbf{f}_{\mathbf{a}, \mathbf{b}}^{(1,0)}, \text{ for } i > 0, \end{aligned} \quad (3.67)$$

where the $\mathbf{f}_{\mathbf{a}, \mathbf{b}}^{(1,0)}$ is reduced by (3.66). As expected, the additional term arises from the remaining terms $\mathbf{R}_{\mathbf{a}, \mathbf{b}}^{(0,0)}$ that emerge during its reduction. Obviously, if we choose these $\frac{4i}{\pi} e^{\pi \text{Im}[\nu]} (-k_{i;1})^{-2\nu_{i;1}-1} f_{\mathbf{a}_i, \mathbf{b}_j}^{(-2\nu_{i;1})}$ in remaining terms (3.65) to be master integrals of this sub-sector, the dependence of the original sector on the sub-sector in the differential equation matrix is

$$\begin{aligned} \Omega_{(\mathbf{a}, \mathbf{b})(\mathbf{c}_i, \mathbf{d}_j)} &= -i \sum_{\mathbf{c}_i} \left(T_n^{-1} \cdot \tilde{\Omega}_{0;1} \cdot T_n\right)_{\mathbf{ac}} \delta_{\mathbf{bd}} \delta_{c_i(1-d_j)} (-1)^{c_i+1} \\ &\quad - i \sum_{\mathbf{d}_j} \left(T_n^{-1} \cdot \tilde{\Omega}_{0;2} \cdot T_n\right)_{\mathbf{bd}} \delta_{\mathbf{ac}} \delta_{d_j(1-c_i)} (-1)^{d_j+1} \\ &= -i \left(T_n^{-1} \cdot \tilde{\Omega}_{0;1} \cdot T_n\right)_{\mathbf{a}(\mathbf{c}_i; 1-b_j)} \delta_{b_j d_j} (-1)^{b_j} \\ &\quad - i \left(T_n^{-1} \cdot \tilde{\Omega}_{0;2} \cdot T_n\right)_{\mathbf{b}(\mathbf{d}_j; 1-a_i)} \delta_{a_i c_i} (-1)^{a_i} \end{aligned} \quad (3.68)$$

where for the subscript of $\Omega_{(\mathbf{a}, \mathbf{b})(\mathbf{c}_i, \mathbf{d}_j)}$, $\mathbf{a}, \mathbf{b}$ are the indices for original sector and $\mathbf{c}_i, \mathbf{d}_j$ are indices for the sub-sector. Then, it is again d log-form DE. Also notice that $\nu_{i;1} = \nu_{j;2}$, $k_{i;1} = k_{j;2}$, so the master integral $\frac{4i}{\pi} e^{\pi \text{Im}[\nu]} (-k_{i;1})^{-2\nu_{i;1}-1} f_{\mathbf{a}_i, \mathbf{b}_j}^{(-2\nu_{i;1})}$ will be consistent for ;1 and ;2. For more than one $G_{\pm\pm}$, the discussion is similar.

4 Summary and Outlook

In this paper, we define the dS integral family, which naturally incorporates the case of time derivatives interaction. We generalize IBP method to dS integral family with respect to $d\tau_i$ and dk_i by (3.4) and (3.5), whose integrands involve special functions. With (3.5) and result of IBP reduction, people also can construct DE with respect to ∂_{k_i} of dS integrals. We indicate the factorization of IBP relations of vertex in the dS correlator. For the vertex integral families, we derive an universal iterative reduction formula for arbitrary n -fold Hankel vertex function families, along with the d log-form DE satisfied by the MI we selected, as listed in (3.50) and (3.55). And the remaining terms come from $G_{\pm\pm}$ are also discussed. Since the tree-level dS correlators only involve integrals over τ_i , we have obtained the reduction and DE of arbitrary tree-level diagrams equivalently.

This paper in fact has presented an alternative pathway toward systematically and efficiently computing dS correlators. Once we have IBP relations, drawing from the experience in flat spacetime, the number of integrals people need to compute will be significantly reduced. Once we have DE, with proper boundary conditions, the remaining steps do not differ from the case of flat spacetime, and numerical result can be efficiently obtained via numerical DE method. Its effectiveness has been validated in flat spacetime, and many existing packages designed for flat spacetime can also be readily applied. For example, Kira could solve IBP reduction of user-defined system [81], AMFlow [73] and DiffExp [74] can numerically solving DE with given boundary condition and DE.

This paper also suggest many interesting open questions. We merely list a part of them as follows to inspire people's future research. Firstly, while we mainly focus on the IBP linear system, how to give a boundary condition of DE is beyond the scope of this paper and haven't been discussed. One can work like the flat space DE, select a simple boundary and analytically or numerically compute it. But whether there is a boundary-determine method like AMFlow [63, 66] in flat cases needs more consideration. For example, one may consider the boundary $k_0 \rightarrow -i\infty$ in dS cases, which shrink all vertex integrals to zero. Secondly, AdS correlator could be considered in the future and the relation of dS and AdS integral could be examined in the perspective of IBP and DE. Thirdly, follow the idea that reclassifying Feynman integrals itself as special function via DE [82], to bring dS integrals closer to a so called analytical result, people need to develop systematical method to analyze this integral family. For example, people may consider how to distinguish the different signal and background in the dS correlator using DE formalism like people have done using partial MB transformation [36] in the future.

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